



विद्याविनियोगाद्विकासः

Project Course Proposal

Term 5

Academic Project

On

Expiry Day Dynamics & VWAP Settlement Anomalies

Area

FINANCE

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INTRODUCTION

Indian derivatives markets present unique microstructure challenges, particularly around expiry-day settlement mechanics. NSE revised its weekly expiry day convention in September 2025; prior to this change, Thursday was the standard expiry day for index and stock derivatives, and this study uses 2022 data, when the Thursday-expiry convention was in effect, and accordingly focuses on NSE expiry Thursdays. During the final 30 minutes of trading on these expiry days, the Volume-Weighted Average Price (VWAP) cash settlement mechanism creates structural incentives for market manipulation. We hypothesize that institutional constraints and basis risk cause textbook arbitrage to break down on the NSE, and this proposal is designed to test that hypothesis empirically.

By utilizing granular NSE historical tick-level order and trade data, this project explores whether the settlement calculation leads to abnormal cash-SSF (Single Stock Futures) basis volatility and pricing errors. The study focuses on understanding if this divergence is driven primarily by institutional and algorithmic execution strategies (often referred to as "settlement punching").

OBJECTIVES OF THE STUDY:

- **How does the true minute-by-minute VWAP trajectory of cash equities deviate from the traded price of the FUTSTK derivative in the final 30 minutes of expiry days?**
- **Which market participants (Proprietary, Custodian, or NCNP) predominantly drive the expiry day volumes and basis divergence?**
- **What is the measurable impact of algorithmic trading on late-day volatility?**
- **Can we detect abnormal order book pressure, such as spikes in cancellations (spoofing) or execution aggressiveness (IOC orders), leading up to the VWAP calculation?**

PROJECT SCOPE

This project will analyze the "CASH" and "FAO" market segments to detect settlement anomalies. To align with the 12-week project timeline, the scope covers:

- **Timeframe Scope:** The 12 monthly expiry Thursdays of 2022, benchmarked against the 12 preceding Wednesdays (serving as a non-expiry control group).
- **Intraday Scope:** The final 30 minutes of the trading session (3:00 PM – 3:30 PM) with CLOB constructed based on all intraday orders.

- **Asset Scope:** A sample of 10 Nifty 50 constituents and their corresponding stock futures the top 5 most liquid names and 5 relatively illiquid names by average daily traded volume to enable a liquidity-contrast analysis of basis divergence and settlement-window behavior, while keeping total data volume manageable.

DATA SOURCES AND METHODOLOGY

For secondary research, we will refer to the NSE Historical Data Orders and Trade Layout Document, review existing literature on derivatives, market microstructure, and settlement manipulation, and consult on how margin requirements affect futures contract rollover. For primary research, we will use NSE Historical Order Data and Trade Data (fixed-record length .DAT.gz files) for the CASH and FAO segments, parsed programmatically in Python and Apache Flink to filter target parameters such as Transaction Time in Jiffies.

ANALYSIS

We will reconstruct the tick-by-tick VWAP using Transaction Price, Transaction Qty, and microsecond-level Transaction Time to track basis fractures across the cash and futures markets. We will use the Client Identity Flag to profile and compare trading behavior across market participants during the settlement window, and the Algo Indicator to segment order flow into Non-Algo and Algo (including SOR) to quantify the role of automated trading in settlement punching. We will analyze the Activity Type field (Order Entry, Modification, Cancellation) in the Orders files to detect abnormal cancellation patterns, and use the Market/Limit Order Indicator and IOC Indicator to measure execution aggressiveness in the final minutes of the session. Finally, we will compare Order Qty against Disclosed Qty to examine the use of iceberg orders during the settlement window as compared to the control windows.

DELIVERABLES

The project will produce a structured overview of VWAP settlement mechanics and baseline basis behavior on the NSE, a report profiling algorithmic and proprietary desk behavior during the settlement window, an interactive Excel model detailing liquidity, execution friction, and basis divergence, and a final research report presenting findings on whether the NSE closing settlement procedure exhibits systematic pricing errors.

PROPOSED TIMELINE OF WORK

Checkpoints	Timeline
Literature Review & Data Extraction Setup	Week 2
Data Parsing & Cleaning Pipeline Setup	Week 4
VWAP & Basis Trajectory Analysis	Week 6
Examination of Participant & Algo Flags	Week 8
Fundamental Conclusions and discussion on the Final Report	Week 10
Final Report & Model Submission	Week 12

Note: Fortnightly update will be sent to the project guide(s)